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| <b>Joel Andrew Miller</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  | (905) 933-7920   joelmiller0430@gmail.com   joelandrewmiller.com |                      |
| M.E.Sc. Machine Learning Engineer specializing in physiological signal processing and AI for human-machine systems. Experienced in building wearable sensor pipelines, biometric data analysis, and deploying ML models to interpret human behaviour in high-stakes environments.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                                                                  |                      |
| <b>WORK EXPERIENCE:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                                                  |                      |
| <b>Machine Learning Engineer – Biological Signals</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  | <b>ITPS Canada</b>                                               | Feb 2026 – Present   |
| <ul style="list-style-type: none"><li>Develop and validate ML models to detect indicators of pilot stress and elevated cognitive workload from multimodal signals (ECG, Pupil Data, flight logs), enabling behavioural assessment in high-stakes flight environments.</li><li>Designed and built a structured flight database to organize and query data by pilot, sortie type, and in-flight events, enabling systematic extraction of behavioural insights to inform human factors research and system design.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                                                                  |                      |
| <b>R&amp;D Aerospace Software Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  | <b>ITPS Canada</b>                                               | Feb 2025 – Feb 2026  |
| <ul style="list-style-type: none"><li>Built a helmet-mounted display project from scratch, including an outside-in optical tracking pipeline, and co-designing in-house display.</li><li>Designed all software for head tracking and sensor-fusion algorithms, achieving ~2° spatial accuracy in cockpit environments.</li><li>Implemented a Kalman filter to correct IMU drift and maintain stable, long-duration tracking under high-motion and vibration conditions.</li><li>Architected and implemented the distributed system enabling fully wireless operation suitable for aircraft with ejection seats.</li><li>Engineered and optimized the UDP networking layer to ensure high-frequency, low-jitter pose transmission, achieving ~3 ms end-to-end latency.</li><li>Designed and executed calibration workflows to align helmet, optical, and aircraft coordinate frames for consistent head-pose reconstruction.</li><li>Collaborated directly with pilots, test engineers, program stakeholders, and leadership to translate operational needs into clear technical requirements and deliverables.</li></ul> |  |                                                                  |                      |
| <b>R&amp;D Engineering Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  | <b>NRC Canada</b>                                                | Sept 2023 – Dec 2023 |
| <ul style="list-style-type: none"><li>Researched and designed a pipeline to transform real-world sensor data (LiDAR, camera, GPS) collected from a vehicle to simulated environments for autonomous vehicle development and testing.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                                                                  |                      |
| <b>Software Engineer/DevOps Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | <b>IBM</b>                                                       | May 2021 – Sept 2022 |
| <ul style="list-style-type: none"><li>Developed automated testing frameworks for IBM’s Order Management System, reducing testing time by 90% and improving system reliability.</li><li>Configured a Dockerized virtual machine integrated with a web application to execute and monitor automated test cases.</li><li>Mentored and onboarded incoming interns, supporting team ramp-up and knowledge transfer.</li><li>Built frontend components to improve user interaction and experience.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |                                                                  |                      |
| <b>EDUCATION:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |                                                                  |                      |
| <b>M.E.Sc Software Engineering with Specialization in Artificial Intelligence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  | <b>Western University</b>                                        | 2023 – Dec 2024      |
| <ul style="list-style-type: none"><li>Employed signal processing, data fusion, time series analysis, and deep learning to model and understand driver’s cognitive behaviour through their physiological signals (ECG, EDA, RSP).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                                                                  |                      |
| <b>B.E.Sc Software Engineering with Co-op</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  | <b>Western University</b>                                        | 2018 – 2023          |
| <b>PROJECTS and LEADERSHIP:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                                                                  |                      |
| <b>Peer Reviewed Studies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |                                                                  |                      |
| <ul style="list-style-type: none"><li><b>Published:</b> “Navigating the Handover: Reviewing Takeover Requests in Level 3 Autonomous Vehicles” (IEEE Open Journal of Vehicular Technology): A comprehensive review of Takeover Requests (TORs) in autonomous vehicles, emphasizing safe human-centered transitions from automation to manual control.</li><li><b>Published:</b> “At the Heart of Intersections: Analyzing Their Influence on Driver Heart Behaviour” (Springer Data Science for Transportation): Investigated intersections’ impact on driver heart rates to inform safer urban traffic designs and autonomous vehicle technologies.</li><li><b>Published:</b> “Modeling Driver State and Takeover Behavior in Conditional Automation Using Physiological Signals” (IEEE): Applied ML to ECG, EDA, and RSP signals to classify takeover occurrence and timing with 76% and 75% accuracy with a small amount of data (&gt;30 samples), advancing adaptive, human-centric autonomous vehicle systems.</li></ul>                                                                                             |  |                                                                  |                      |
| <b>CARLA Reinforcement Learning Agent (<i>Masters Project</i>)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                                                                  |                      |
| <ul style="list-style-type: none"><li>Designed and implemented a reinforcement learning agent for autonomous vehicle navigation in the CARLA simulation environment, leveraging Deep Q-Learning and PPO algorithms.</li><li>Fine-tuned reward functions to balance efficiency and exploitation, improving learning rates and decision-making accuracy.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                                                                  |                      |
| <b>FoodSnap (<i>Capstone Project</i>)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |                                                                  |                      |
| <ul style="list-style-type: none"><li>Engineered a machine learning application to analyze the nutritional content of meals from single images, integrating image segmentation, depth estimation networks, and the USDA API.</li><li>Achieved accurate identification of food types and volume estimation, enabling precise calorie and macronutrient calculations.</li><li>Demonstrated the project’s scalability and potential impact in nutrition tracking through real-world testing scenarios.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                                                  |                      |
| <b>Western AI Club (<i>Project Manager</i>)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                                                                  |                      |
| <ul style="list-style-type: none"><li>Led a team of five developers to design, train, and deploy an iOS image classification app for waste categorization, promoting sustainable disposal practices.</li><li>Implemented transfer learning techniques to optimize the model, achieving 98% accuracy in classifying waste types.</li><li>Oversaw project milestones, ensuring timely delivery and seamless integration of app features.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                                                                  |                      |
| <b>Skills:</b> Python, PyTorch, Tensorflow, scikit-learn, pandas, NumPy, OpenCV, SQL, signal processing, ECG/EDA/RSP analysis, wearable sensors, sensor fusion, Kalman filtering, IMU processing, camera calibration, optical tracking, anomaly detection, machine learning, computer vision, reinforcement learning, real-time systems, UDP networking, distributed systems, Docker, Git, human-machine interfaces, Linux.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |                                                                  |                      |